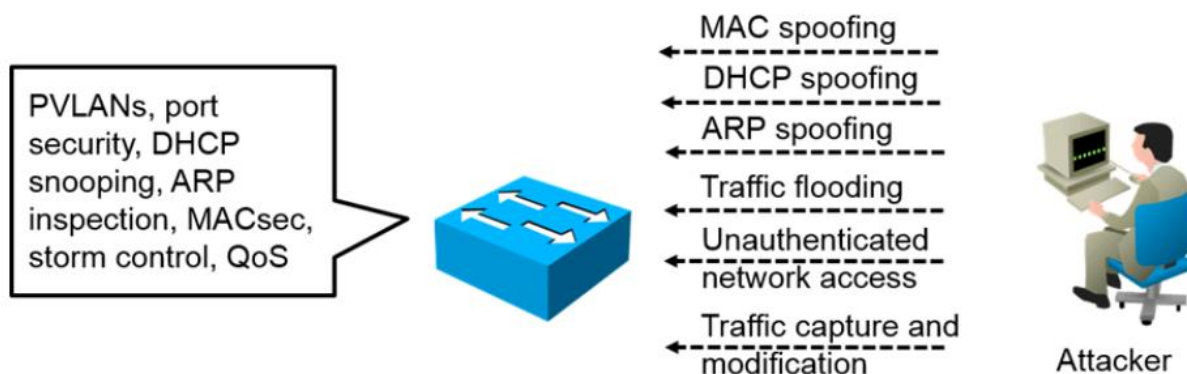


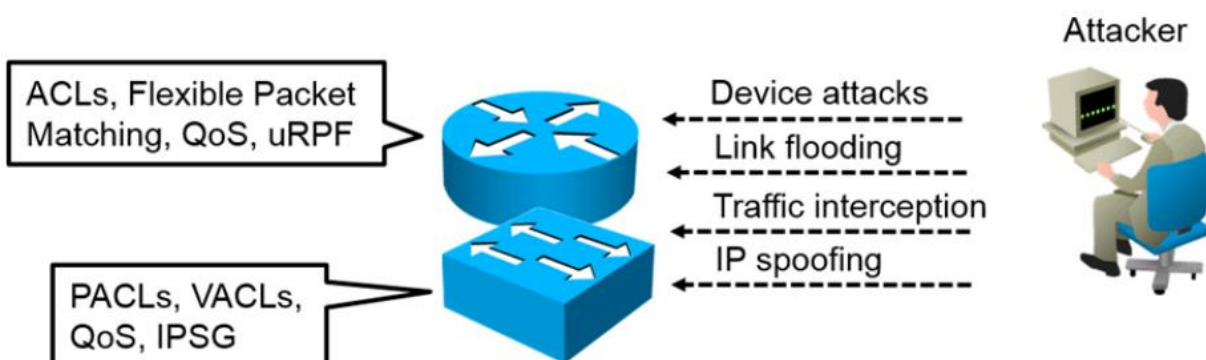
Data Plane:

- o Data Plan traffic is a traffic, which is from the **user to the user**.
- o Data plane traffic that is "just passing through" to get to other destinations.
- o Traffic that is being forwarded through the network also called transit traffic.
- o Example user sending traffic from one part of network to access a server in another part.
- o A failure in the data plane results in the customer's traffic not being able to be forwarded.

Layer 2 Data Plane Security Controls



L3 Data Plane Security Controls



Best Practices for Protecting the Data Plane.

- o Create and enable Access Control List on network devices, Routers, Switches.
- o Create and deploy VLANs, Private VLAN (Virtual Local Area Network) in Network.
- o Enable Spanning Tree Protocol (STP) guard such as BPDU Guard and Root Guard.
- o Enable if possible, Inter Operating System (IOS) IPS (Intrusion Prevention System).
- o Create and enable the Zone-based Firewall feature on Cisco Routers to protect it.
- o Apply and Enable TCP Intercept feature which protect the device from a SYN-flood.
- o Apply and Enable Unicast Reverse Path Forwarding mitigate spoofed IP packets.
- o Port security to protect against MAC address flooding and CAM overflow attacks.
- o Enable the DHCP snooping feature to prevent a rogue DHCP attack in the network.
- o Enable Dynamic ARP inspection (DAI)to protect against ARP spoofing, ARP poisoning.